

Software Engineering Lab 1
Group Travel Expense Simplification Engine
Requirements Engineering
Domain: Retail, E-Commerce & Finance
Actors: Group Traveler, Trip Admin, Daily Exchange
Rate Service

Submitted by:

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Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compute the optimal debt settlement graph using min-flow algorithms to minimize total peer-to-peer payment transactions.	High	Pass: Group balance settles to net zero with the minimum possible payment edges. Fail: Sum of debts does not equal sum of credits or unnecessary payment edges remain.	Minimizes the number of transactions required to settle group expenses.
FR-002	Functional	The system shall allow a Group Traveler to record an expense with the amount, currency, payer, and participating travelers.	High	Pass: A valid expense is stored with all required details and associated travelers. Fail: The expense is rejected if any required detail is missing or invalid.	Expense recording is necessary to calculate each traveler's share and balance.
FR-003	Functional	The system shall convert expenses recorded in foreign currencies into the selected settlement currency using the applicable daily exchange rate.	High	Pass: The converted amount matches the recorded foreign amount multiplied by the applicable daily exchange rate. Fail: The system uses an unavailable or incorrect exchange rate.	Enables accurate expense sharing when group expenses use multiple currencies.
FR-004	Functional	The system shall calculate the net balance of each traveler based on the expenses paid and the traveler's share of group expenses.	High	Pass: The system correctly identifies each traveler as owing or being owed the calculated net amount. Fail: The resulting balances do not reconcile to zero across the group.	Individual net balances are required before generating the settlement plan.

FR-005	Functional	The system shall display the optimized settlement plan showing the payer, recipient, and amount for each required transaction.	High	Pass: Every required settlement transaction displays a payer, recipient, and amount and collectively settles all group balances. Fail: Any transaction is missing or the displayed transactions do not settle the balances.	Allows travelers to clearly understand how their outstanding expenses should be settled.
NFR-001	Nonfunctional	The settlement calculation engine shall execute in under 100 ms for groups of up to 50 travelers with 500 expenses.	High	Pass: Benchmarking confirms settlement calculation completes in less than 100 ms for 50 travelers and 500 expenses. Fail: Execution time reaches or exceeds 100 ms.	Ensures the settlement engine responds quickly for moderately large groups.
NFR-002	Nonfunctional	The system shall protect stored expense and settlement data from unauthorized access by enforcing authenticated access for all group expense and settlement operations.	High	Pass: Unauthenticated users cannot access, create, modify, or view group expense and settlement data. Fail: Any protected operation is accessible without authentication.	Protects potentially sensitive financial information shared within the travel group.